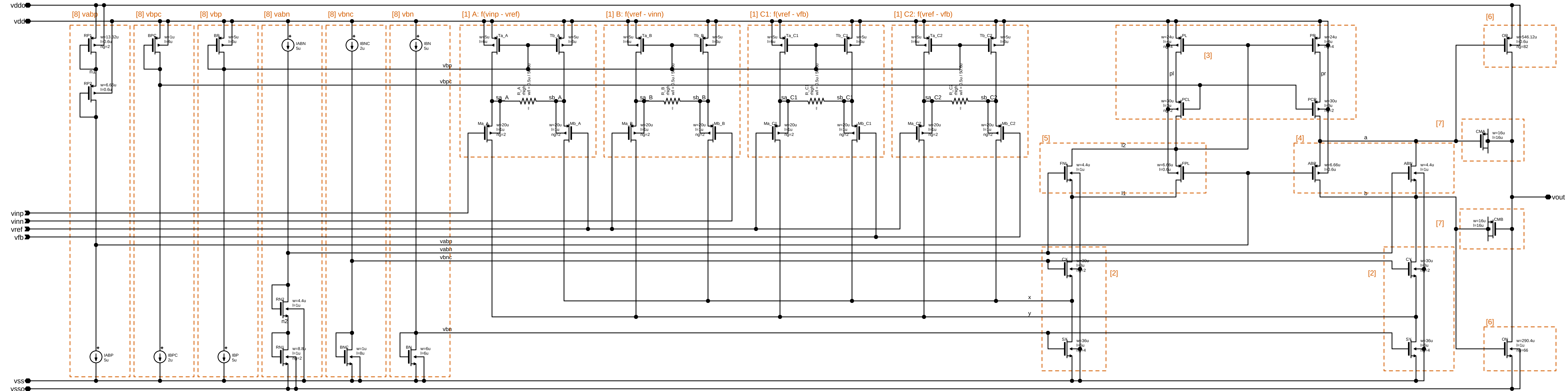


d2s_mpdda_bias_flat: d2s_mpdda and d2s_bias_lp in one sheet, all at transistor level

same devices as d2s_mpdda_biased (d2s_mpdda + d2s_bias_lp); unit devices and nets carry the unit name: Ta_A, Tb_A, R_A, Ma_A, Mb_A, sa_A, sb_A, ... rhigh body = vss

the bias nets are internal here (d2s_mpdda_biased keeps them as pins for CACE); frozen defaults lcas = 3, wc = lc = 16u, rl = 50.6u, iab = 5u, ibnc = 2u; block descriptions: README.md



[1] DDA units A, B, C1, C2: split-tail PMOS pair, rhigh between the sources. Each adds $f(gp - gn)$ to the fold nodes: the gp device drains to y, the gn device to x.

A: $v_{inp} - v_{ref}$, B: $v_{ref} - v_{inn}$, C1 = C2: $v_{ref} - v_{fb}$. The loop forces $f(v_{inp} - v_{ref}) + f(v_{ref} - v_{inn}) = 2 f(v_{fb} - v_{ref})$, so $v_{fb} - v_{ref} = (v_{inp} - v_{inn})/2$.

[2] fold: sinks SX, SY (gate vbn) and NMOS cascodes CX, CY (gate vbnc). x and y are the fold nodes.

[3] PMOS cascoded mirror: PL, PCL is the input side (gates tied to I2), PR, PCR the output side into node a. Cascode gates vbp.

[4] class-AB control: ABP (gate vabp) and ABN (gate vabn) between a and b. With RP1/RP2 and RN1/RN2 in the bias it sets the quiescent current of OP and ON.

[5] FPL, FNL: copy of [4] in the mirror input branch (between I2 and I1), so both fold branches carry the same element.

[6] output devices OP (gate a), ON (gate b), drains on vout.

[7] Miller capacitors CMA (between a and vout) and CMB (between vout and b): hv PMOS in accumulation, gate over n-well, 16u x 16u.

[8] bias: ideal reference currents into diode-connected devices.

vbp: BP 5u/6u at 5 uA, for the unit tails Ta, Tb (5u/6u, 5 uA each). vbn: BN 6u/6u at 5 uA, for the sinks SX, SY (36u/6u, 30 uA each).

vbp: BPC 1u/4u at 2 uA, for the mirror cascodes PCL, PCR. vbnc: BNC 1u/8u at 2 uA, for the fold cascodes CX, CY.

vabp: RP2 (= ABP) on RP1 (2 x ABP) at 5 uA, for ABP, FPL. vabn: RN2 (= ABN) on RN1 (2 x ABN) at 5 uA, for ABN, FNL.